Sr. No	Name of the Laboratory	Number of students per set up(Batch Size)	Name of the Important Equipment	Weekly utilization status(all the courses for which the lab is utilized)	Technical Manpower Support		
					Name of the Technical staff	Designation	Qualification
1	Engineering Gr	75	Different types	05	Different types	Jr. Technical St	Diploma in Mec
2	Fluid Mechanic	50	.Bernoulli's The	03	Chhote Lal Mal	Jr. Technical St	Diploma in Mec
3	Dynamics of M	50	1. Motorised G	03	Anit Mahato	Jr. Technical St	Diploma in Aut
4	Mechanical En	50	1. Universal Te	03	Jitendra Kumar	Jr. Technical St	Diploma in Mec
5	Heat Transfer L	50	1.Thermal Con	03	Ramesh Munda	Jr. Technical St	Diploma in Mec
6	Internal Combu	50	1. Computerise	03	C.K.Singh,	Technical Supe	Diploma in Aut
7	Mechanical En	50	1. Insight Solar	03	Anit Mahato, R	Jr. Technical St	Diploma in Aut
8	Robotics &Autc	50	1. Car Simulato	03	Jitendra Kumar	Jr. Technical St	Diploma in Mec
9	Automobile En	50	1. Cut Section	03	C.K.Singh,	Technical Supe	Diploma in Aut
10	Refrigeration &	50	1.Cooling using	03	Ashok Kumar (	Jr. Technical St	Diploma in Mec
11	Computer Aide	50	70+ computers	05	Pramod Kumar	Technical Supe	BCA

7.2 Additional Facilities Created for Improving the Quality of Learning Experience in Laboratories (20)

Sr. No	Name of the Facility	Details	Purpose for creating facility	Utilization	Relevance to POs/PSOs
1	Department Computer Lab (Location: Main Building – Ground Floor – Mechanical Engg,)	70+ modern desktop computers with high RAM and dedicated GPUs. Licensed software: AutoCAD, Solidworks, Ansys, MATLAB etc.	To enable UG, PG and PhD students to perform simulation, data analysis, design, and modelling relevant to the curriculum and research.	Laboratories are actively used by faculty for funded research projects, Final-year UG, PG and PhD students utilize lab resources for research and project Simulation.	PO1, PO2, PO3, PO4, PO5, PO10, PSO1, PSO3
2	Reverse Engineering Lab. (Location: Main Building – Ground Floor – Mechanical Engg,)	The lab utilizes virtual machines, specialized software or 3D scanning for product innovation, and CAD model generation from physical parts.	To enable UG, PG and PhD students to perform simulation, data analysis, design, and experiment relevant to the curriculum and research.	Laboratory is actively used by faculty for funded research projects, Final-year UG, PG and PhD students utilize lab resources for research and project Simulation.	PO2, PO3, PO4, PO5, PO12, PSO1
3	Computational Lab. (Location: Main Building – Ground Floor – Mechanical Engg.)	15+ modern desktop computers with high RAM and dedicated GPUs. Licensed software: Ansys Fluent, MATLAB, etc	To enable UG, PG and PhD students to perform simulation, data analysis, design, and modelling relevant to the curriculum and research.	Laboratory is actively used by faculty for funded research projects, Final-year UG, PG and PhD students utilize lab resources for research and project Simulation.	PO1, PO2, PO4, PO5, PO10, PSO1, PSO3
4	Wind tunnel (Location: I.C. engine lab. – Ground Floor – Mechanical Engg.)	The apparatus is used for generating a controlled, uniform air stream to test the aerodynamic forces (lift, drag) on models, using a blower/fan, settling chamber for laminar flow, a contraction cone, and a test section.	To enable UG, PG and PhD students to perform experimental analysis for testing aerodynamic forces (lift, drag) on models, using a blower/fan.	Laboratories are actively used by faculty for funded research projects, Final-year UG, PG and PhD students utilize lab resources for research and project Simulation.	PO1, PO2, PO4, PO7, PSO1
5	Universal Testing machine (Location: Main Building – Ground Floor – Mechanical Engg., Autonomous Lab)	Tensile Testing up to 10000 kg Bending Testing Compression Testing up to 10000 kg	To enable UG, PG and PhD students to perform experimental analysis for testing tensile strength of various samples.	Equipment is actively used by faculty for funded research projects, Final-year UG, PG and PhD students utilize for their research project.	PO1, PO2, PO4, PO5, PSO1
6	High Temperature Air Jet Erosion (Location: Main Building – Ground Floor – Mechanical Engg., Strength of Material Lab)	Up to 140 m/s velocity of erodent, up to a temperature of 900 ^{o}C. Can test erosion behaviour of materials at different attack angles.	For UG projects and funded research projects.	Used for UG students undertaking projects and internships and PhD students.	PO1, PO2, PO4, PO7, PSO1
7	Experimental Module for Stick-slip Frictional Vibration (Location: Main Building – Ground Floor – Mechanical Engg., Dynamics of Machine Lab)	Stick-slip frictional vibration of a circular disc sliding on another rigid rotating disc can be analysed.	To enable PG and PhD students to perform experimental analysis of vibration induced due to the friction in circular discs.	Equipment is actively used by faculty for funded research projects, Final-year UG, PG and PhD students utilize for their research project.	PO1, PO2, PO4, PSO1

8	Biogas plant (Location: Behind Hostel 11, Mesra campus)	From 250 kg/day of organic waste (leftover food/vegetables) converts into 35 to 40 m3 of biogas from the CSRT digester, used for cooking activities in hostel 11.	Develop Bio gas production process from organic waste.	Plant is actively used by faculty for funded research projects, Final-year UG, PG and PhD students utilize for their research project. Summer Internship is undertaken by internship students.	PO6, PO7, PO8, PO11, PSO2
9	Scratch tester (Location: Main Building – Ground Floor – Mechanical Engg., Strength of Material Lab)	Finding scratch hardness and critical load of failures of materials and coatings.	For UG projects and funded research projects.	Used for UG students undertaking projects and internships and PhD students.	PO1, PO2, PO4, PSO1
10	TIERA Machine Fault Simulator (TMFS) (Location: Main Building – Ground Floor – Mechanical Engg., Dynamics of Machine Lab)	It is used to study the signature of common machinery faults, such as unbalance, alignment, resonance bearing, rotor dynamics ,gearbox, belt drive, mechanical rub, induction motor, pump, compressor and fan.	To enable UG, PG, and PhD students to perform experimental analysis for testing faults in machinery.	Equipment is actively used by faculty for funded research projects, Final-year UG, PG and PhD students utilize for their research project.	PO1, PO2, PO4, PO5, PO11, PSO2
11	Friction stir welding machine IC Engine Lab – Mechanical Engg.	Friction stir welding (FSW) machines are solid-state joining systems that use a rotating, non-consumable tool to generate frictional heat and soften materials, creating high-strength, low-distortion welds without melting.	To enable UG, PG, and PhD students to perform experiment on friction stir welding.	Equipment is actively used by faculty for funded research projects, Final-year UG, PG and PhD students utilize for their research project.	PO1, PO2, PO3, PO4, PO5, PSO1, PSO2

7.3 Maintenance of laboratories and overall ambiance (10)

7.3 Maintenance of Laboratories and Overall Ambiance

Laboratory Maintenance Overview – Department of Mechanical Engineering

All laboratories in the Department of Mechanical Engineering are well-maintained to ensure smooth academic and research activities.

- In each laboratory, a Professor-in-Charge and a technician are assigned for Lab maintenance. Their responsibility is to look over the overall regular maintenance of the equipment, and keep all records related to maintenance, repairs and purcha
- A Laboratory Safety & Maintenance Committee oversees the overall maintenance and functionality of the laboratories across the department.
- Minor issues, such as equipment malfunctions or electrical problems, are promptly attended to by the departmental technical staff, often in coordination with the Central Electricity Department and the Central Workshop.
- For major breakdowns, the respective equipment manufacturers are contacted, and formal work orders are issued for repairs.
- For sophisticated analytical instruments, Annual Maintenance Contracts (AMCs) are secured at the time of procurement. Every effort is made to renew these AMCs regularly to ensure uninterrupted operation and minimize downtime.

1. **Daily Maintenance Tasks**
    - Benchtop and Equipment surface Cleaning.
    - Waste disposal (chemical, sharps etc.,) in designated bins.
    - Check water taps, gas valves, and power outlets for leaks or hazards.
2. **Weekly/Monthly Maintenance**
    - Inspect and clean air filters.
    - Check inventory of oil, and consumables
3. **Equipment Maintenance**
    - Follow manufacturer guidelines for preventive maintenance
    - Maintain logbooks for major equipment
4. **Safety Checks**
    - Emergency eyewash and shower stations
    - Check fire extinguishers, first aid kits, and spill kits
    - Check Do’s and Don’t list.
5. **Documentation and Record-Keeping**
    - Maintain SOPs (Standard Operating Procedures)
    - Update maintenance logs for equipment

Laboratory Overall Ambience – Department of Mechanical Engineering

The overall ambience of the Mechanical Engineering laboratories is reasonably good.

- Sufficient number of windows are available to ensure effective ventilation and adequate natural light.
- Lighting system is very effective, along with the natural light in every corner of the rooms.
- An adequate number of ceiling/wall-mounted fans are installed to maintain comfortable working conditions.
- The lab that needs air conditioning is well equipped with an air conditioning system.
- Each laboratory is equipped with cupboards to safely store spare parts, glassware, and laboratory instruments.
- Properly designed and sufficient tables are provided for housing equipment and conducting experiments.
- Adequate seating arrangements, including stools for students, are available for comfortable working.
- Analytical Instrumentation and computer Laboratories are air-conditioned for improved equipment stability and operator comfort.
- All laboratories have access to a consistent and good-quality water supply.
- Generator backup is available in all laboratories to ensure uninterrupted power supply during outages.

7.4 Safety measures in laboratories (10)

Sr. No	Laboratory Name	Safety Measures
1	Engineering Graphics	First-aid kits, Proper use of electric devices, Fire extinguisher Emergency Contact Numbers Displayed, Equipment Safety Instructions, Laboratories are equipped with MCBs, and instructed to always handle drawing instruments and laboratory equipment carefully to prevent accidents and injuries.
2	Fluid Mechanics & Hydraulic Machines Lab	First-aid kits, Proper use of electric devices, and Fire extinguisher, Equipment Safety Instructions, Laboratories are equipped with MCBs, and Do's and Don't list. Student instruct turbines, or hydraulic equipment without proper supervision and following the prescribed operating procedure. This helps prevent accidents, equipment damage, and injuries cau moving machinery.
3	Mechanical Engineering Lab I	First-aid kits, Proper use of electric devices, and Fire extinguisher Emergency Contact Numbers Displayed, Power Backup, Equipment Safety Instructions, Laboratories are equip list. Student instructed to follow the instructions carefully, as the rotating parts/ Sharp edges/ Electrical connections of the Equipment might cause serious damage.
4	Dynamics of Machines Lab	First-aid kits, Proper use of electric devices, Fire extinguisher Emergency Contact Numbers Displayed, Power Backup, Equipment Safety Instructions, Laboratories are equipped Student instructed to avoid touching the rotating parts and avoid loose clothes as the rotating parts of the Equipment might cause serious damage.
5	Heat Transfer Lab	First-aid kits, Proper use of electric devices, and Fire extinguisher Emergency Contact Numbers Displayed, Equipment Safety Instructions, Laboratories are equipped with MCBs instructed to not touch heated surfaces or equipment without proper protective gloves, as they can cause burns. Wear appropriate personal protective equipment (PPE).
6	Internal Combustion Engines Lab	First-aid kits, Proper use of electric devices, and Fire extinguisher Emergency Contact Numbers Displayed, Power Backup, Equipment Safety Instructions, Laboratories are equip list. Student instructed to keep hands, clothing, and loose objects away from moving engine parts such as belts, pulleys, and rotating shafts while the engine is running.
7	Mechanical Engineering Lab II	First-aid kits, Proper use of electric devices, and Fire extinguisher Emergency Contact Numbers Displayed, Equipment Safety Instructions, Laboratories are equipped with MCBs instructed to never touch the rotating parts/ Sharp edges/ Electrical connections of the Equipment, which might cause serious damage.
8	Robotics & Automation Lab	First-aid kits, Proper use of electric devices, and Fire extinguisher Emergency Contact Numbers Displayed, Power Backup, Equipment Safety Instructions, Laboratories are equip list. Student instructed to maintain a safe distance from moving robotic arms and never enter the robot's operating area while it is running.
9	Automobile Engineering Lab	First-aid kits, Proper use of electric devices, and Fire extinguisher Emergency Contact Numbers Displayed, Equipment Safety Instructions, Laboratories are equipped with MCBs instructed to always wear appropriate personal protective equipment (PPE), such as safety goggles, gloves, and safety shoes, while working on vehicles and machinery.
10	Refrigeration & Air Conditioning Lab	First-aid kits, Proper use of electric devices, and Fire extinguisher Emergency Contact Numbers Displayed, Equipment Safety Instructions, Laboratories are equipped with MCBs instructed to handle refrigerants carefully and ensure proper ventilation to avoid inhalation of refrigerant gases and prevent leakage hazards.

11	Computer Aided Design & Drafting Lab	First-aid kits, Proper use of electric devices, and Fire extinguisher Emergency Contact Numbers Displayed, Power Backup, Equipment Safety Instructions, Laboratories are equip list. Student instructed to use computers and electrical equipment properly and avoid touching electrical connections with wet hands to prevent electric shock.

7.5 Project laboratory/research laboratory /centre of excellence (20)

Sl No.	Name of the Laboratory
1	Research and Project lab. (Scratch tester , High temperature air jet erosion) (Location: Main building, Strength of Material Lab)
2.	Reverse Engineering Lab. (3-d printer) (Location: Main Building – Ground Floor – Mechanical Engg,)
3.	Computational Lab. (Various simulation software, Matlab, Ansys etc) (Location: Main Building – Ground Floor – Mechanical Engg.)
4.	Robotics Lab. (PID-CASCAD controller, UTM machine) (Location: Main Building – Ground Floor – Mechanical Engg., Robotics &Automation Lab)
5.	Computer Aided Design & Drafting Lab. (Location: Main Building – Ground Floor – Mechanical Engg.)
6.	Internal Combustion Engines Lab. (Wind tunnel, Friction stir welding machine) (Location: IC Engine Lab. building, Ground Floor – Mechanical Engg.)
7.	Dynamics of Machines Lab. (TIERA Machine Fault Simulator (TMFS), stick-slip frictional vibration machine) (Location: Main Building – Ground Floor – Mechanical Engg.)
8.	Biogas plant (Location: Behind Hostel 11, Mesra campus)

8 CONTINUOUS IMPROVEMENT (80)

8.1 Actions taken based on the results of evaluation of each of the COs, POs & PSOs (40)